

TABLE I

CWRU BEARING DATASETS—OPERATING CONDITIONS, LABELS, AND CLASS-WISE SAMPLE DISTRIBUTION (60/20/20 TRAIN/VAL/TEST SPLIT, SEED 42). ALL DATASETS: $T_{\text{WIN}} = 170.7 \text{ ms}$, $\Delta f = 11.7 \text{ Hz/BIN}$, JENSEN-SHANNON $D_{\text{JS}} \leq 0.064$. BF: BALL FAULT; IF: INNER RACE FAULT; OF: OUTER RACE FAULT @6'. FAULT DIAMETERS: 007, 014, 021 MIL.

Dataset	Class	Train	Val	Test
CWRU 12k DE	BF007	234	117	118
	BF014	236	117	118
	BF021	235	118	118
	IF007	236	117	119
	IF014	234	117	117
	IF021	235	117	118
	OF007	236	118	117
	OF014	234	118	118
	OF021	236	118	118
	Normal	708	473	471
	Total	2824	1530	1532
CWRU 12k FE	BF007	233	117	117
	BF014	235	118	117
	BF021	233	117	117
	IF007	234	117	117
	IF014	234	117	117
	IF021	234	117	117
	OF007	235	117	117
	OF014	69	22	22
	OF021	69	22	22
	Normal	708	473	471
	Total	2484	1337	1334
CWRU 48k DE	BF007	234	118	58
	BF014	234	59	117
	BF021	234	117	58
	IF007	234	58	117
	IF014	236	117	14
	IF021	235	118	58
	OF007	235	58	117
	OF014	175	117	118
	OF021	236	59	118
	Normal	175	117	117
	Total	2228	938	892

TABLE II

CLASS-WISE SAMPLE DISTRIBUTION OF THREE CWRU BEARING DATASETS (60/20/20 TRAIN/VAL/TEST SPLIT, SEED 42). ALL DATASETS: $T_{\text{WIN}} = 170.7 \text{ ms}$, $\Delta f = 11.7 \text{ Hz/BIN}$, JENSEN-SHANNON $D_{\text{JS}} \leq 0.064$. BF: BALL FAULT; IF: INNER RACE FAULT; OF: OUTER RACE FAULT @6'. FAULT DIAMETERS: 007, 014, 021 MIL.

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	IF007	234	117	117
	IF014	234	117	117
	IF021	234	117	117
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	OF007	235	58	117
	OF014	175	117	118
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	Normal	175	117	117
	Total	2228	938	892

TABLE III

CLASS-WISE SAMPLE DISTRIBUTION OF THREE CWRU BEARING DATASETS (60/20/20 TRAIN/VAL/TEST SPLIT, SEED 42). ALL DATASETS: $T_{\text{WIN}} = 170.7 \text{ ms}$, $\Delta f = 11.7 \text{ Hz/BIN}$, JENSEN-SHANNON $D_{\text{JS}} \leq 0.064$. BF: BALL FAULT; IF: INNER RACE FAULT; OF: OUTER RACE FAULT @6'. FAULT DIAMETERS: 007, 014, 021 MIL.

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	IF021	235	117	118
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	OF021	236	118	118
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	BF014	234	59	117
	BF021	234	117	58
	IF007	234	58	117
	IF014	236	117	14
	IF021	235	118	58
	OF007	235	58	117
	OF014	175	117	118
	OF021	236	59	118
	Normal	175	117	117
	Total	2228	938	892

TABLE IV
PERFORMANCE OF THREE DATASETS IN THE CLOSED-SET SETTING, INCLUDING ACCURACY (ACC%) AND F1-SCORE (F1%)

Dataset	Label Num	Flexmatch		BYOL		Flexmatch		SSCL	HSFD	OSCL	
		Acc	F1	Acc	F1	Acc	F1	Acc	Acc	Acc	F1
CWRU	10	72.40±11.85	66.05±14.93	93.30±0.03	92.56±0.03	65.27±4.77	54.50±3.83	64.75	66.73	97.26±0.09	97.05±0.17
	50	84.38±15.63	81.59±19.73	94.88±0.04	94.34±0.04	80.56±7.37	76.44±10.04	88.92	79.81	97.28±0.08	97.02±0.09
	100	85.23±15.31	81.32±19.72	96.23±0.04	96.16±0.18	89.21±7.16	81.66±9.38	91.53	92.13	97.49±0.04	97.25±0.04
	150	87.80±6.66	84.55±13.36	97.27±0.04	97.09±0.05	92.07±7.65	91.79±0.41	95.77	94.01	97.42±0.05	97.18±0.05
	250	91.88±1.79	90.85±2.20	97.78±0.03	97.55±0.03	92.07±4.02	90.83±4.88	97.10	97.97	97.44±0.02	97.20±0.02
	500	95.51±8.54	94.73±10.06	97.49±0.05	97.23±0.05	93.40±5.71	92.25±6.65	99.03	99.29	99.12±0.09	99.03±0.10
	2000	99.18±0.87	99.09±0.97	98.39±0.06	98.23±0.07	96.33±2.66	94.55±5.95	99.97	99.84	99.37±0.11	99.30±0.13
MFPT	3	84.63±7.96	79.65±13.65	85.64±0.64	82.71±0.58	84.85±7.64	80.15±10.58	85.73	87.38	96.85±0.08	95.76±0.10
	9	90.56±4.46	88.40±4.30	92.19±0.46	89.78±0.61	92.10±7.45	90.08±9.03	92.09	91.33	96.83±0.07	95.74±0.08
	30	93.06±3.67	91.16±4.77	92.55±0.06	89.02±0.09	93.67±6.56	93.48±6.33	93.36	91.69	97.11±0.05	96.11±0.07
	60	94.83±1.98	93.55±2.42	91.77±0.05	88.81±0.07	94.24±3.46	92.35±4.70	94.97	94.21	97.15±0.07	96.19±0.11
	300	97.38±1.99	96.44±2.70	98.37±0.24	97.86±0.31	95.63±5.64	95.23±5.56	98.98	99.01	99.07±0.11	94.95±0.15
	600	97.77±1.21	97.54±1.07	98.25±0.08	97.68±0.11	97.12±2.53	96.37±2.05	99.72	99.60	99.26±0.06	99.00±0.08
XJTU-SY	15	40.16±4.65	35.77±6.78	69.84±0.11	68.57±0.08	34.13±3.02	26.98±4.35	—	—	92.27±0.23	91.04±0.25
	45	55.92±14.76	53.55±15.43	72.17±0.08	70.27±0.11	69.01±3.36	58.92±3.30	—	—	92.17±0.33	90.86±0.58
	150	78.85±7.34	76.61±9.10	80.46±0.50	79.26±0.57	79.88±7.71	88.10±6.87	—	—	93.32±0.18	92.04±0.19
	300	80.36±6.54	78.44±8.06	84.80±0.51	83.46±0.63	90.24±3.87	89.61±4.25	—	—	92.92±0.23	91.71±0.19
	1500	89.93±1.91	89.44±1.10	92.56±0.35	91.48±0.51	89.17±4.52	87.85±3.55	—	—	93.34±0.07	93.18±0.14
	3000	91.92±2.25	87.70±2.26	91.43±0.12	90.51±0.10	90.41±9.10	89.61±0.69	—	—	94.29±0.11	93.02±0.12
HIT	4	79.53±9.79	79.43±9.48	57.91±1.74	57.43±1.76	64.03±7.36	61.27±4.6	—	—	93.05±0.93	92.95±0.96
	12	93.14±0.19	93.04±0.17	79.41±0.21	79.44±0.25	75.96±2.65	72.55±4.27	—	—	94.88±0.25	94.83±0.26
	40	95.70±1.55	95.70±1.55	86.53±0.88	86.83±1.05	93.21±1.66	93.24±1.66	—	—	95.17±0.06	95.12±0.06
	80	90.75±3.76	90.98±3.35	88.26±0.88	87.94±0.93	91.76±1.23	91.63±1.23	—	—	94.99±0.16	94.95±0.17
	400	99.26±0.32	99.26±0.33	97.48±0.95	96.91±0.64	98.01±0.93	98.02±0.87	—	—	99.76±0.03	99.76±0.05
	800	99.87±0.04	99.87±0.04	99.55±0.03	99.56±0.03	98.52±1.2	98.51±1.22	—	—	99.85±0.06	99.86±0.07

TABLE V
PERFORMANCE ON CWRU DATASET IN CLOSED-SET SETTING
(ACCURACY & F1-SCORE)

TABLE VII
PERFORMANCE ON XJTU-SY DATASET IN CLOSED-SET SETTING
(ACCURACY & F1-SCORE)

	Label	Flexmatch		BYOL		Flexmatch		SSCL	HSFD	OSCL		BYOL	Flexmatch
		Acc	F1	Acc	F1	Acc	F1	Acc	Acc	Acc	F1	Acc	Acc
10	72.40±11.85	66.05±14.93	93.30±0.03	92.56±0.03	65.27±4.77	54.50±3.83	64.75	66.73	97.26±0.09	97.05±0.17	93.30±0.03	34.13±3.02	26.98±4.35
50	84.38±15.63	81.59±19.73	94.88±0.04	94.34±0.04	80.56±7.37	76.44±10.04	88.92	79.81	97.28±0.08	97.02±0.09	94.88±0.04	69.01±3.36	58.92±3.30
100	85.23±15.31	81.32±19.72	96.23±0.04	96.16±0.18	89.21±7.16	81.66±9.38	91.53	92.13	97.49±0.04	97.25±0.04	96.23±0.04	79.88±7.71	88.10±6.87
150	87.80±6.66	84.55±13.36	97.27±0.04	97.09±0.05	92.07±7.65	91.79±0.41	95.77	94.01	97.42±0.05	97.18±0.05	97.27±0.04	89.61±4.25	87.85±3.55
250	91.88±1.79	90.85±2.20	97.78±0.03	97.55±0.03	92.07±4.02	90.83±4.88	97.10	97.97	97.44±0.02	97.20±0.02	97.78±0.03	90.41±9.10	89.61±0.69
500	95.51±8.54	94.73±10.06	97.49±0.05	97.23±0.05	93.40±5.71	92.25±6.65	99.03	99.29	99.12±0.09	99.03±0.10	97.49±0.05	90.41±9.10	89.61±0.69
2000	99.18±0.87	99.09±0.97	98.39±0.06	98.23±0.07	96.33±2.66	94.55±5.95	99.97	99.84	99.37±0.11	99.30±0.13	99.09±0.97	90.41±9.10	89.61±0.69

TABLE VI
PERFORMANCE ON MFPT DATASET IN CLOSED-SET SETTING
(ACCURACY & F1-SCORE)

TABLE VIII
PERFORMANCE ON HIT DATASET IN CLOSED-SET SETTING (ACCURACY & F1-SCORE)

	Label	Flexmatch		BYOL		Flexmatch		SSCL	HSFD	OSCL		BYOL	Flexmatch
		Acc	F1	Acc	F1	Acc	F1	Acc	Acc	Acc	F1	Acc	Acc
3	84.63±7.96	79.65±13.65	85.64±0.64	82.71±0.58	84.85±7.64	80.15±10.58	85.73	87.38	96.85±0.08	95.76±0.10	85.64±0.64	64.03±7.36	61.27±4.6
9	90.56±4.46	88.40±4.30	92.19±0.46	89.78±0.61	92.10±7.45	90.08±9.03	92.09	91.33	96.83±0.07	95.74±0.08	92.19±0.46	75.96±2.65	72.55±4.27
30	93.06±3.67	91.16±4.77	92.55±0.06	89.02±0.09	93.67±6.56	93.48±6.33	93.36	91.69	97.11±0.05	96.11±0.07	92.55±0.06	93.21±1.66	93.24±1.66
60	94.83±1.98	93.55±2.42	91.77±0.05	88.81±0.07	94.24±3.46	92.35±4.70	94.97	94.21	97.15±0.07	96.19±0.11	91.77±0.05	91.76±1.23	91.63±1.23
300	97.38±1.99	96.44±2.70	98.37±0.24	97.86±0.31	95.63±5.64	95.23±5.56	98.98	99.01	99.07±0.11	94.95±0.15	98.37±0.24	98.01±0.93	98.02±0.87
600	97.77±1.21	97.54±1.07	98.25±0.08	97.68±0.11	97.12±2.53	96.37±2.05	99.72	99.60	99.26±0.06	99.00±0.08	97.77±1.21	98.52±1.2	98.51±1.22

TABLE IX
SUMMARY OF PREVIOUS WORKS (IN CHRONOLOGICAL ORDER) THAT CONSIDER SIMILAR ACOUSTIC SIGNAL DATASETS.

Ref.	Year	Method	Input feature
[?]	2019	AE	Mel spectrogram
[?]	2020	IDNN	Mel spectrogram
[?]	2020	FCN	MDF
[?]	2020	One-shot learning	Spectrogram
[?]	2021	Fully connected U-Net	Mixed features, i.e., MFCC, chroma feature, Mel spectrogram, spectral contrast, and Tonnetz
[?]	2021	CNN	Mel spectrogram
[?]	2021	Convolutional VAE	Mel spectrogram
[?]	2021	ResNet, Gaussian mixture models, and One-class SVM	Mel spectrogram
[?]	2021	SVM and MLP	Mel spectrogram
[?]	2021	RawdNet	Mel spectrogram

TABLE X
COMPARISON RESULTS UNDER GAUSSIAN NOISE INTERFERENCE

Method	Original	Noise interference level		
	Data	Slight	Medium	Strong
Proposed method	99.90% (4.30%)	98.09% (34.94%)	95.29% (52.49%)	92.28% (66.41%)
Ensemble TICNN [?]	99.50%	99.61%	98.22%	82.05%
LDR-CNN [?]	100%	99.82%	99.45%	98.88%
Deep Transfer Learning [?]	97.95%	96.72%	94.68%	90.32%
AAAnet [?]	99.48%	99.83%	99.83%	98.28%
MCNN-LSTM [?]	98.46%	88.19%	81.41%	77.27%